APPENDIX 3

Some important formulae for Integral

(1) $\int dx = x$; $\frac{d}{dx}(x) = 1$

(2) $\int x^n dx = \left(\frac{x^{n+1}}{n+1}\right)$; $\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$

(3) $\int cudx = c\int udx$ where c is a constant

(4) $\int (u \pm v \pm w)dx = \int udx \pm \int vdx \pm \int wdx$

(5) $\int \frac{1}{x}dx = \ln x + C$

(6) $\int e^x dx = e^x + C$

(7) $\int \cos\theta d\theta = \sin\theta + C$

(8) $\int \sin\theta d\theta = -\cos\theta + C$

Some important formulae in Differential calculus

1. $\frac{d}{dx}(c) = 0$, if c is a constant

2. If y = cu, where c is a constant and u is a function of x then
$$\frac{dy}{dx} = \frac{d}{dx}(cu) = c\frac{du}{dx}$$

3. If y = u ±v ±w where u,v and w are functions of x then
$$\frac{dy}{dx} = \frac{d}{dx}(u \pm v \pm w) = \frac{du}{ds} \pm \frac{dv}{dx} \pm \frac{dw}{dx}$$

4. If $y = x^n$, where n is the real number then
$$\frac{dy}{dx} = \frac{d}{dx}(x^n) = nx^{n-1}$$

5. If y = uv, where u and v are functions of x then
$$\frac{dy}{dx} = \frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

6. If y is a function of x, then $dy = \frac{dy}{dx}.dx$

7. $\frac{d}{dx}(e^x) = e^x$

8. $\frac{d}{dx}(\ln x) = \frac{1}{x}$

9. $\frac{d}{d\theta}(\sin\theta) = \cos\theta$

10. $\frac{d}{d\theta}(\cos\theta) = -\sin\theta$

11. If y is a trigonometric function of θ and θ is the function of t, then
$$\frac{d}{dt}(\sin\theta) = \cos\theta\frac{d\theta}{dt}$$
$$\frac{d}{dt}(\cos\theta) = -\sin\theta\frac{d\theta}{dt}$$

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